

Probability Paradigm	Statistical Paradigm
Sample Space	Event
Union	Intersection
Complement	Mutually Exclusive
Axioms of Probability	Conditional Probability

The population and its parameters are unknown. We can make probabilistic statements about the population given the information that is observed in the sample.	The population and its parameters are assumed to be known. We can compute the probability of drawing a sample with certain characteristics.
A subset of outcomes of the sample space	The set of all possible outcomes, S, of a random process.
• For two events, the intersection consists of all outcomes that are contained in both of the two events. • Denoted $E_1 \cap E_2$	• For two events, the union consists of all outcomes that are contained in either of the two events. • Denoted $E_1 \cup E_2$
• Two events are mutually exclusive if they have no outcomes in common • $E_1 \cap E_2 = \emptyset$	• The complement of an event is the set of outcomes in the sample space that are not in the event. • Denoted E' or E^c
A probability in which we condition on some event $P(A B) = \frac{P(A \cap B)}{P(B)} \text{ given } P(B) \neq 0$	1. $P(S) = 1$ 2. $0 \leq P(E) \leq 1$ 3. For two events E_1 and E_2 with $E_1 \cap E_2 = \emptyset$, $P(E_1 \cup E_2) = P(E_1) + P(E_2)$

Independence	Bayes' Theorem
Random Variable	Joint PMF
Marginal PMF	Conditional PMF
Joint PDF	Marginal PDF
Conditional PDF	Covariance

$P(B_i A) = \frac{P(A B_i)P(B_i)}{P(A)} = \frac{P(A B_i)P(B_i)}{\sum_{j=1}^n P(A B_j)P(B_j)}$	Two events are independent of each other if conditioning on one does not affect the probability of the other. $P(A B) = P(A)$
• $f_{X,Y}(x,y) = P(X=x, Y=y)$ • $\sum_X \sum_Y f_{X,Y}(x,y) = 1$	• A function that associates a unique numerical value with each outcome in the sample space. • Random variables are denoted with capital letters, and realizations of random variables are denoted with lowercase letters.
• $f_{X Y}(x y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$ • $f_{Y X}(y x) = \frac{f_{X,Y}(x,y)}{f_X(x)}$	• $f_X(x) = \sum_Y f_{X,Y}(x,y)$ • $f_Y(y) = \sum_X f_{X,Y}(x,y)$
• $f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)dy$ • $f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)dx$	• $P((X,Y) \in \mathcal{R}) = \int \int_{\mathcal{R}} f_{X,Y}(x,y)dx dy$ • $\int \int f_{X,Y}(x,y)dx dy = 1$
$\text{Cov}(X,Y) = \sigma_{XY} = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)]$ $= \mathbb{E}[XY] - \mu_X \mu_Y$	• $f_{X Y}(x y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$ • $f_{Y X}(y x) = \frac{f_{X,Y}(x,y)}{f_X(x)}$

Correlation	Properties of Covariance
Linear Combinations of Random Variables	NOTE:
Propogation of Error Formulas	Sample Mean
Sample Variance	p^{th} quantile
Median Absolute Deviation (MAD)	Constructing a Histogram

1. $\text{Cov}(X, X) = \text{Var}(X)$ 2. The covariance and correlation only measure linear association 3. If X and Y are independent, then the covariance is 0. However, zero covariance does not necessarily imply that X and Y are independent.	$\rho = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}$
A linear combination of normal random variables is itself normal	Let $X_1, \dots, X_n$ be independent random variables and let $a_0, a_1, \dots, a_n$ be constants. Also let $U = a_0 + a_1 X_1 + \dots + a_n X_n.$ $\mathbb{E}[U] = a_0 + a_1 \mathbb{E}[X_1] + \dots + a_n \mathbb{E}[X_n]$ $\text{Var}(U) = a_1^2 \text{Var}(X_1) + \dots + a_n^2 \text{Var}(X_n)$
$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$	Let $U = f(X_1, \dots, X_n)$ where f is not linear. Then • $\mathbb{E}[U] = f(\mathbb{E}[X_1], \dots, \mathbb{E}[X_n])$ • $\text{Var}(U) = \left(\frac{\partial f}{\partial x_1}\right)^2 \text{Var}(X_1) + \dots + \left(\frac{\partial f}{\partial x_n}\right)^2 \text{Var}(X_n)$
$\tilde{x}_p = \tilde{x}_{\left(\frac{i-0.5}{n}\right)} = x_{(i)}$	$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$
1. Divide the range of the data into non-overlapping intervals of equal length. A rough rule is to choose $\sqrt{n}$ intervals. 2. Determine how many observations fall into each interval. 3. Center the rectangle over the midpoint of each interval. 4. The height of a rectangle is the frequency of observations in that interval (or proportion of total)	$\text{MAD} = \text{median} \{ x_1 - \tilde{x} , \dots, x_n - \tilde{x} \}$

Quantile Plot	Quantile-Quantile (QQ) Plot
Theoretical QQ Plot/Probability Plot (PP Plot)	Kernel Density Estimator (KDE)
Boxplot	Sample Correlation
Sampling Distribution	Sampling Distribution of $\bar{X}$
Central Limit Theorem (CLT)	Normal Approximation to Binomial Distribution

For two sets of data with respective quantile function, $\tilde{x}_{p,1}$ and $\tilde{x}_{p,2}$, the QQ plot is a graph of the ordered pairs $(\tilde{x}_{p,1}, \tilde{x}_{p,2})$ for the appropriate values of p	A plot of $\tilde{x}_{i-0.5/n}$ (on the y-axis) versus p (on the x-axis). For the ordered data, plot the points $(\frac{i-0.5}{n}, x_{(i)})$, and connect the points with straight line segments.
Let $(x_1, \dots, x_n)$ be a random sample of size n from some univariate distribution with density $f(\cdot)$. The density $f(\cdot)$ can be approximated by the KDE $\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - x_i}{h}\right),$ where $K(\cdot)$ is a kernel function and h is the bandwidth.	Same as the QQ plot, except that a theoretical distribution (typically the normal distribution) is plotted on one of the axes.
Estimates the strength of the linear association between the two variables. $r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 (y_i - \bar{y})^2}}$	• Lower line of box is Q_1, middle line of box is Q_2, and upper line of box is Q_3. • The upper whisker extends to the largest data point less than or equal to $Q_3 + 1.5 \times \text{IQR}$. The lower whisker extends to the smallest data point greater than or equal to $Q_1 - 1.5 \times \text{IQR}$. • Points lying outside of the whiskers are marked with asterisks (and are potential outliers).
$\bar{X} \sim (\mu, \sigma^2/n)$	The probability distribution of a statistic that describes how the statistic varies across all possible samples of the same size selected from the population of interest.
• $\hat{p} = X/n \sim N\left(p, \frac{p(1-p)}{n}\right)$ • $X \sim N(np, np(1-p))$ So long as $np \geq 10$ and $n(1-p) \geq 10$	Let $X_1, \dots, X_n$ be a random sample from a distribution with mean μ and variance σ^2. As $n \rightarrow \infty$, the sampling distribution of $\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \rightarrow N(0, 1)$ meaning it converges to a standard normal distribution. Thus, if n is sufficiently large, then the distribution of $\bar{X}$ is approximately normal with mean μ and variance σ^2/n.

Sampling Distribution of S^2	Point Estimator
Point Estimate	Bias
Variance (of an Estimator)	Mean Squared Error (MSE)
Relative Efficiency (RE)	Parametric Bootstrap
Nonparametric Bootstrap	Method of Moments (MoM) Estimation

• The point estimator of a population parameter θ is a function of the sample data $X_1, \dots, X_n$, and is labeled $\hat{\theta}$. • It is a random variable.	$\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$
• Defined as $\mathbb{E}[\hat{\theta}] - \theta$ • An estimator is unbiased if $\mathbb{E}[\hat{\theta}] = \theta$	The numerical value of $\hat{\theta}$ calculated from the observed sample data, $X_1 = x_1, \dots, X_n = x_n$.
$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + (\text{Bias}(\hat{\theta}))^2$	$\text{Var}(\hat{\theta}) = \mathbb{E}[\hat{\theta}^2] - \mathbb{E}[\hat{\theta}]^2$
1. Take random sample of size n from $f(x \theta)$ 2. From random sample, estimate parameter $\hat{\theta}_1^*$ 3. Repeat steps 2 and 3, B times to obtain $(\hat{\theta}_1^*, \dots, \hat{\theta}_n^*)$ 4. $\bar{\theta}^* = (1/B) \sum_{i=1}^B \hat{\theta}_i^*$ 5. $S_{\theta} = \sqrt{\frac{1}{B-1} \sum_{i=1}^B (\hat{\theta}_i^* - \bar{\theta}^*)^2}$	The relative efficiency of an estimator $\hat{\theta}_2$ to estimator $\hat{\theta}_1$ is defined as $\text{RE} = \frac{\text{MSE}(\hat{\theta}_1)}{\text{MSE}(\hat{\theta}_2)}$
Let $X_1, \dots, X_n$ be a random sample from $f(x \theta_1, \dots, \theta_m)$ where $\theta_1, \dots, \theta_m$ are parameters whose values are unknown. The moments estimators $\hat{\theta}_1, \dots, \hat{\theta}_m$ are obtained by equating the first m sample moments to the corresponding population moments and solving for $\theta_1, \dots, \theta_m$ $\mathbb{E}[X^k] = (1/n) \sum_{i=1}^n X_i^k, \quad k = 1, \dots, m$	1. Treat the n observations from the sample as the population 2. Take a random sample of size n WITH REPLACEMENT from the data 3. From this random sample, estimate the parameter $\hat{\theta}_1^*$ 4. Follow steps 3-5 of the parametric bootstrap

Two Disadvantages with MLEs	Invariance Property of MLEs
Large Sample Property of MLEs	Prior Distribution (for Bayes Estimation)
Posterior Distribution	Bayes Estimator
NOTE:	Confidence Level
Confidence Interval	Confidence Interval for Single Mean μ (Known σ)

Let $\hat{\theta} = (\hat{\theta}_1, \dots, \hat{\theta}_m)$ be the MLEs of the parameters $\theta = (\theta_1, \dots, \theta_m)$. Then the MLE of any function of θ, $h(\theta_1, \dots, \theta_m)$ is the function $h(\hat{\theta}_1, \dots, \hat{\theta}_m)$.	1. Using them requires that you know which underlying distribution the sample comes from 2. Calculus based methods sometimes fail when trying to derive the MLEs, so numerical methods must be used to find them
The prior distribution of the parameter θ is its probability distribution, denoted $f(\theta)$.	Under some general conditions on the joint distribution of the sample, when n is large, 1. The MLE of any parameter θ is approximately unbiased, $\mathbb{E}[\hat{\theta}] \approx \theta$ 2. The variance of the MLE is nearly as small as can be achieved by any estimator 3. The MLE has an approximate normal distribution
The mean of the posterior distribution denoted $\tilde{\theta}$	$f(\theta x_1, \dots, x_n) = \frac{f(x_1, \dots, x_n \theta)f(\theta)}{\int_{-\infty}^{\infty} f(x_1, \dots, x_n \theta)f(\theta) d\theta}$
• Gives the percentage of values that will be captured around the point estimate by the sampling distribution of the statistic. • Given by $100(1 - \alpha)\%$ where α is a number between 0 and 1.	Bayes estimators will (usually) outperform MLEs when the prior information is relatively sharp, the data are consistent with the prior, and the sample size is small to moderate
$P\left(-z_{\alpha/2} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq z_{\alpha/2}\right) = 1 - \alpha$	The confidence interval for a parameter θ is the range of values captured between the lower bound θ_L and upper bound θ_U such that $P(\theta_L \leq \theta \leq \theta_U) = 1 - \alpha$

Confidence Interval for Single Mean μ (Unknown σ)	Pivot
CI Interpretation BEFORE Sample is Selected	CI Interpretation AFTER Sample is Selected
NOTE:	Factors that influence width of CI
Independent Samples (Two Population CI for Means)	Paired Samples
Paired Samples CI	Independent Samples (Equal Variance) CI

A quantity whose distribution does not depend on any unknown parameters	$P\left(-t_{\alpha/2, n-1} \leq \frac{\bar{X} - \mu}{S/\sqrt{n}} \leq t_{\alpha/2, n-1}\right) = 1 - \alpha$ $P(\mu \in \bar{X} \pm t_{\alpha/2, n-1}(S/\sqrt{n})) = 1 - \alpha$
• The endpoints of the CI are fixed, known constants • We say that "We are $100(1 - \alpha)\%$ confident that the population parameter lies between the lower and upper bounds."	A level $100(1 - \alpha)\%$ confidence interval is one computed by a method that, in the long run, will contain the population parameter in $100(1 - \alpha)\%$ of all the times that it is used. In repeated samples of size n, this method of constructing intervals will produce confidence intervals in which approximately $100(1 - \alpha)\%$ of the intervals contain the true parameter of interest.
1. Sample size (Controllable) 2. Confidence level (Controllable) 3. Sampling distribution (Uncontrollable) 4. Standard deviation (Uncontrollable)	The probability that we select a sample whose confidence interval contains the true parameter is $1 - \alpha$.
Each values from one sample is matched or paired with another value, and a random selection of paired units is taken.	• The samples from the two populations are not linked to each other • The sample sizes from the populations can differ • Each observation must be independent from all other observations of both samples
The POI is $\mu_X - \mu_Y$. The formula for the CI is $(\bar{x} - \bar{y}) \pm t_{\alpha/2, \nu} \left(s \sqrt{\frac{1}{n_X} + \frac{1}{n_Y}} \right)$ where, $S^2 = \frac{\sum (X_i - \bar{X})^2 + \sum (Y_i - \bar{Y})^2}{n_X + n_Y - 2}$ $\nu = n_X + n_Y - 2$	The POI is μ_d and can be estimated with the statistic $\bar{D} = \frac{1}{n} \sum_{i=1}^n (X_i - Y_i)$ The formula for the CI is: $\bar{d} \pm t_{\alpha/2, n-1} (s_d/\sqrt{n})$ NOTE: Ensure that $n \geq 30$ to have approximate normality by CLT

Independent Samples (Unequal Variance) CI	Satterthwaite's degrees of freedom
CI for a Single Proportion	CI for Two Population Proportions
CI for Single Variance	F Distribution
CI for Two Variances	Prediction Interval (PI)
PI (Single Mean)	Tolerance Interval (TI)

$\nu = \frac{(s_X^2/n_X + s_Y^2/n_Y)^2}{(s_X^4/n_X^2)/(n_X - 1) + (s_Y^4/n_Y^2)/(n_Y - 1)}$	The POI is $\mu_X - \mu_Y$. The formula for the CI is $(\bar{x} - \bar{y}) \pm t_{\alpha/2, \nu} \sqrt{\frac{s_X^2}{n_X} + \frac{s_Y^2}{n_Y}}$ where ν is Satterthwaite's degrees of freedom
POI is $p_1 - p_2$. The formula for the CI is $(\hat{p}_1 - \hat{p}_2) \pm z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}}$ So long as $n_1\hat{p}_1 > 10, n_1(1 - \hat{p}_1) > 10$ and $n_2\hat{p}_2 > 10, n_2(1 - \hat{p}_2) > 10$	$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$
- Let U and V be two independent random variables such that $U \sim \chi_{\nu_1}^2$ and $V \sim \chi_{\nu_2}^2$. Then, $W = \frac{U/\nu_1}{V/\nu_2}$ follows an F distribution with ν_1 numerator df and ν_2 denominator df. F_{ν_1, ν_2} is not the same as F_{ν_2, ν_1}	$\frac{(n-1)s^2}{\chi_{n-1, 1-\alpha/2}^2} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi_{n-1, \alpha/2}^2}$
An interval estimate for an individual observation	POI is σ_X^2/σ_Y^2. The formula for the CI is $\frac{s_X^2}{s_Y^2 (F_{n_X-1, n_Y-1, 1-\alpha/2})} \leq \frac{\sigma_X^2}{\sigma_Y^2} \leq \frac{s_X^2}{s_Y^2 (F_{n_X-1, n_Y-1, \alpha/2})}$
An interval that will contain at least a specified fraction $1 - \gamma$ of all future observations from a given population with some preassigned probability $1 - \alpha$	The $100(1 - \alpha)\%$ PI for X_{n+1} is $\bar{x} \pm t_{\alpha/2, n-1} s \sqrt{1 + \frac{1}{n}}$

Tolerance Interval (Single Mean)	Bonferroni Intervals
Parametric Estimation of CDF	Nonparametric estimation of CDF
Dvoertetzky-Kiefer-Wolfowitz-Massart Bound	Null Hypothesis
Alternative Hypothesis	Type I Error
Type II Error	NOTE:

Let α_s be the simultaneous error desired for m confidence intervals. Then assign $\alpha_i = \alpha_s/m$ to each individual interval	The standard form for a TI of a random sample from a $N(\mu, \sigma^2)$ population is $\bar{x} \pm Ks$ where K is a constant that solves $P\left(\Phi\left(\frac{\bar{X} + KS - \mu}{\sigma}\right) - \Phi\left(\frac{\bar{X} - KS - \mu}{\sigma}\right) \geq 1 - \gamma\right) = 1 - \alpha$
- Estimator: $\hat{F}_X(x) = \frac{\#\{X_i \leq x\}}{n}$ (ECDF) - CI: (See Dvoretzky-Kiefer-Wolfowitz-Massart Bound)	- Estimator: $\hat{F}_X(x) = F_X(x \hat{\theta})$ - CI: $\hat{F}_X(x \hat{\theta}_L) \leq F_X(x) \leq \hat{F}_X(x \hat{\theta}_U)$
What is currently believed to be true	$\left \hat{F}_X(x) - F_X(x) \right \leq \alpha_n = \sqrt{\frac{1}{2n} \log(2/\alpha)} \quad \forall x$
- Rejecting the null hypothesis when the null is true $P(\text{Reject } H_0 H_0 \text{ is true}) = \alpha$	What you set out to show is true
A Type I error is typically more serious than a Type II error. This is why we choose to fix α, the Type I error rate.	- Failing to reject the null hypothesis when the null is false. $P(\text{Fail to Reject } H_0 H_a \text{ is true})$

Test Statistic (TS)	Critical Value
Rejection Region	p -value
Power	Interpreting Power
Shapiro-Wilks Test of Normality	t -test
Paired t -test	Independent t -test (Equal variances)

The quantile from the sampling distribution of the test statistic based on α and the direction of the alternative hypothesis such that is the TS exceeds this value, then we reject the null hypothesis.	Tells you how unusual the data observed are, assuming that the null hypothesis is actually true.
The probability of obtaining data as or more extreme than what was observed given that the null hypothesis is true.	The range of values exceeding the critical value(s) such that is the TS falls in this region, then we reject the null hypothesis.
To interpret power, you assume that the alternative hypothesis is true, and specifically that $\mu = \mu_1$, which is a particular value in the alternative space. If, for example, the power is 0.7356 for a given value μ_1, then you have a 73.56% chance of drawing a sample whose sample mean leads you reject the null hypothesis.	• Defined to be $1 - \beta$ • The probability of correctly rejecting a false null hypothesis
• POI: μ • TS: $\frac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim T_{n-1}$	$x_1, \dots, x_n$ come from a normal distribution $x_1, \dots, x_n$ do not come from a normal distribution
• POI: $\mu_1 - \mu_2$ • TS: $\frac{\bar{X}_1 - \bar{X}_2}{S\sqrt{1/n_1 + 1/n_2}} \sim T_{n_1 + n_2 - 2}$ where S^2 is the pooled variance	• POI: μ_d: mean of differences • TS: $\frac{\bar{X}_d - \mu_d}{S_d/\sqrt{n}} \sim T_{n_d - 1}$

Independent t -test (Unequal Variances)	Test for Single Proportion
Test for Two Proportions	Test for Single Variance
Test for Two Variances	False Discovery Rate (FDR)
Non-parametric Hypothesis Tests	

• POI: p • TS: $\frac{\hat{p}-p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} \sim N(0, 1)$ • Conditions: $np_0 \geq 10, n(1-p_0) \geq 10$	• POI: $\mu_1 - \mu_2$ • TS: $\frac{\bar{X}_1 - \bar{X}_2}{\sqrt{S_1^2/n_1 + S_2^2/n_2}} \sim T_\nu$ where ν is Satterthwaite's df
• POI: σ^2 • TS: $\frac{(n-1)S^2}{\sigma_0^2} \sim \chi_{n-1}^2$ • Assumption: $X_1, \dots, X_n \sim N(0, 1)$	• POI: $p_1 - p_2$ • TS: $\frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})(1/n_1 + 1/n_2)}} \sim N(0, 1)$ • $\hat{p} = \frac{n_1\hat{p}_1 + n_2\hat{p}_2}{n_1 + n_2}$
The mean of the number of false rejections divided by the number of rejections	• POI: σ_1^2/σ_2^2 • TS: $\frac{S_1^2}{S_2^2} \sim F_{n_1-1, n_2-1}$
	• Sign Test • Wilcoxon Signed Rank Test